

STAT231 (Data Science) Project Description

The last major deliverable for STAT231 will provide an opportunity for you to undertake a data science project on a topic of your choice. The project is an chance to show off what you've learned. It is a major component of the class and a great way to integrate what you've learned. Your final output for this project will be: 1) a technical report written in Quarto; and 2) an oral presentation delivered to the class (sometimes accompanied by a Shiny app).

Group Formation

You may work individually or in a group of two students: you may choose your own group.

Assignment

Your task is to use data to tell us something interesting. This project is deliberately open-ended to allow you to fully explore your creativity. There are two main guidelines that must be followed:

1. Your project must be centered around data. Preferably, you will work with large, complex, and/or messy data. The more challenging your data set is, the better. Some things that will make your data science project more ambitious and more interesting are:
 - Combining two or more data sets that are not obviously related. For example, an analysis of tweets predicting elections paper would need to combine data that had nothing to do with politics with a data set of congressional districts, and both data sets had to be spatially-aware.
 - Accessing a live data stream. There are many APIs on the Internet that allow you to access data that changes on a continual basis. This is in contrast to static data that was collected at some point in time and does not change. If your project can respond dynamically to a live data stream, then this will make it more interesting.
 - Creating a sophisticated dynamic visualization using multiple elements of Shiny.

Please note that the above stipulations are not requirements – they are merely examples of considerations that may make your data more challenging to work with and your insights more effective – and thus your project more ambitious.

2. Your project must **tell us** something. On one extreme are *data art* projects like Memo Akten's *Forms*, which involve no statistical analysis. On the other extreme are *data wrangling* projects, which may involve minimal visualization but generate some tangible result. Your project can be anywhere on this spectrum, but expectations may be different depending on where you are on the scale. An example of a project that doesn't tell us anything, would be something that downloads a single data source and summarizes it, with some perfunctory visualization.

General Rules You may discuss your project with other students, but each of you will have a different topic, so there is a limit to how much you can help each other. You may consult other sources, but you must appropriately cite and credit these sources in your report, presentation, and Shiny app. Feel free to consult with me at any time.

1 Components

1.1 Roster

Your roster is due by Google form submission (link provided on Moodle) by the end of the day on **Friday, March 29th**.

1.2 Proposals

Your initial proposal is due by the end of the day on **Tuesday, April 2nd** (via Google form). I will provide feedback on **Wednesday, April 3rd**. The final proposal will be due on **Tuesday, April 9th** (submitted via a new repo that I will create for your group).

Prior to writing your initial proposal you should get together with your groupmate and brainstorm a few ideas before coalescing around one idea. Be ambitious! Once you decide on a topic that interests you, think about what you would like to end up with as a final result—without worrying about how to get there. Try to visualize what your end product will look like. Will it be an interactive map? A Shiny application? Don't think about coding, or a particular data set, or what you know how to do now: I will be available to help serve as a resource. If you come up with something ambitious and original, you'll be more motivated to learn new things as you go in order to accomplish your goal.

Content Your initial proposal will be submitted via a Google form. The revised proposal should be submitted using the GitHub repo I will set up for the projects. The revised proposal will be included in a Quarto file called 'report.qmd' (with associated pdf) and should contain the following content:

1. Group Members: List the members of your group (or just you)
2. Title: The title of your project
3. Purpose: Describe the general topic/phenomenon you want to explore, as well some carefully considered questions that you hope to address. You should make an argument *motivating* your work. Why should someone be interested in what you are doing? What do you hope people will learn from your project?
4. Data: As best you can, describe where you will find your data, and what *kind* of data it is. Will you be working with spatial data in shapefiles? Will you be accessing an API to a live data source? Be as specific as you can, listing URLs and file formats if possible.
5. Variables: List, and briefly describe, each variable that you plan to incorporate. If you can, be specific about units, scale, etc.
6. End Product: Describe what you hope to deliver as a final product. Will it be a Shiny application that will be posted on the Internet (I anticipate that most groups will incorporate some form of dynamic visualization)? Will it be a GoogleMaps mash-up? Will it be an R package that provides an API to a live data source? Will it be an exploration of an interesting network dataset?

1.3 Data submission

Please close the issue for the ‘data submission’ deliverable by the end of the day on **Tuesday, April 23rd** to confirm that your data are complete.

1.4 Update

Please respond to the following questions posted as an issue on your project GitHub repo by midnight on **Tuesday, April 30th**.

1. Are any additional datasets needed? If not fully complete, describe any issues that you are having, and what your plan is for getting the rest of the data.
2. What is the single biggest unresolved issue you are having? Please describe it briefly, and what your plan is for resolving this issue.
3. What is the next thing that you plan to work on (and when do you plan on working on it)?

1.5 Draft report and slides

An initial draft of your project report and your initial slides is due by the end of the day on **Friday, May 3rd**.

1.6 Presentation

An effective oral presentation is an integral part of this project. One of the objectives of this class is to give you experience conveying the results of a technical investigation to a non-technical audience in a way that they can understand. Whether you choose to stay in academia or pursue a career in industry, the ability to communicate clearly is of paramount importance. As a data scientist, the burden of proof is on you to convince your audience that what you are saying is true. If your audience (who may very well be less knowledgeable about statistics than you are) cannot understand your results or their interpretations, then the technical merit of your project is irrelevant.

On **Thursday, May 9th** from 1-3pm in Webster you or your group will make a 10 minute oral presentation to the class. We will be running a very tight schedule. You should prepare effective and creative slides. Your goal should be to convey to your audience a clear understanding of your topic, along with a basic understanding of your project, and how well it addresses the question you posed. You should **not** tell us everything that you did, or show a bunch of things that you tried that didn’t work well. After hearing your talk, each student in the class should be able to answer:

1. What was your project about?
2. What was your data like, and what techniques did you apply to it?
3. What were your findings?

You should prepare electronic slides for your talk. Use your creativity! Google Slides are fine. If your project has any interactive elements, please show them off for us! Demonstrating using a Shiny app can be effective. One thing you should *not* do is walk us through your calculations or code in RStudio unless there’s something compelling there.

Groups will **submit their slides via GitHub by midnight of the day *before* your presentation** (as a pdf) along with a link to your Shiny app (if you are creating one).

Advice There are many sources of advice for how to make a good presentation, but an excellent place to start is:

<http://techspeaking.denison.edu/>

Watch the videos on this site to identify some common mistakes. Here is some general advice:

- Budget your time. You have at most 10 minutes, and we will be running a very tight schedule. If your talk runs too short or too long, it makes you seem unprepared. Rehearse your talk ahead of time (with your group) several times in order to get a better feel for your timing (I will make myself available for rehearsals). Note also that you may have a tendency to talk faster during your actual talk than you will during your rehearsal. Talking faster in order to speed up is not a good strategy—you are much better off simply cutting material ahead of time. You will probably have a hard time getting through 25 slides in 10 minutes.
- Don't write too much on each slide. You don't want people to have to read your slides, because if the audience is reading your slides, then they aren't listening to you. You want your slides to provide visual cues to the points that you are making—not substitute for your spoken words. Concentrate on graphical displays and bullet-pointed lists of ideas.
- Put your problem in context. Remember that most of your audience will have little or no knowledge of your subject matter. The easiest way to lose people is to dive right into technical details that require prior domain knowledge. Spend a few minutes at the beginning of your talk introducing your audience to the most basic aspects of your topic and present some motivation for what you are studying.
- Speak loudly and clearly. Remember that you know more about your topic than anyone else in the room, so speak and act with confidence!
- Tell a story—not necessarily the whole story. It is unrealistic to expect that you can tell your audience everything that you know about your topic in 10 minutes. You should strive to convey the big ideas in a clear fashion, but not dwell on the details. Your talk will be successful if your audience is able to walk away with an understanding of what your research question was, how you addressed it, and what the implications of your findings are.

1.7 Write-Up

Your final write-up is due by the end of the day on **Monday, May 13th**: please let me know that you are ready for me to grade by closing the issue that I created. (Extensions are possible if requested in advance.)

In your write-up, you should tell a data science audience about your project, why they should care about it, and what you have discovered. Your audience will be people like you—current or aspiring data scientists. Keep in mind that this audience is extraordinarily diverse in terms of skills and abilities, so you should assume very little about what they might know. However, your audience is reasonably tech-savvy, so you need not “dumb-down” your analysis.

Your write-up should make it clear to me and any other student in the class what methods and techniques you have used to produce your finished product.

Content You should not present *all* of the R code that you wrote throughout the process of working on this project. However, the write-up should contain the *minimal* set of R code that is necessary to generate your data, carry out your wrangling, fit various models, and understand your results and findings. If you make a claim, it *must* be justified by explicit calculation. A knowledgeable reviewer should be able to compile your `.qmd` file without modification, and verify every statement that you have made. All of the R code necessary to produce your figures and tables *must* appear in the write-up.

Motivation Be sure to motivate your topic at the beginning of your write-up. You should try to hook the reader early on. Assume that your audience is a skeptical data scientist who has stumbled across your blog but has very little time to read it. Can you give them a reason to continue reading? A cool visualization or result can help.

Format You don't need to follow a specific format in the write-up, but you should start with an introductory paragraph and finish with a conclusion. These paragraphs need not follow the formal writing style that you would use in most other classes. Here, a colloquial style that is accessible to a lay reader is appropriate.

Nevertheless, your write-up should address the following questions:

1. Why should anyone care about this?
2. What is this about? Do *not* assume that your readers have any domain knowledge! The burden of explanation as to what you are talking about is on you! For example, if your project involves phylogenetic trees, do not assume that your audience has anything other than a basic, lay understanding of genetics.
3. Where did your data come from? What kind of data was it? Is there a link to the data or some other way for the reader to follow up on your work?
4. What are your findings? What kind of statistical computations (if any) have you done to support those conclusions? Again, while the R code will show you performing the calculation, it is up to you to interpret, in English sentences, the results of these calculations. Do not forget about units, axis labels, etc.
5. What are the limitations of your work? Be clear so that others do not misinterpret your findings. To what population do your results apply? Do they generalize? Could your work be extended with more data or computational power or time to analyze? How could your study be improved? Suggesting plausible extensions don't weaken your work – they strengthen it by connecting it to future work.

Style The Quarto format is designed to be an interactive document (not dissimilar to a blog entry). Take advantage of this by including hyperlinks, figures, videos, etc. to provide context for the reader. Use Quarto elements like links, lists, $\text{\LaTeX}$, and images as needed. Include a bibliography that includes citations for your data and key packages that you are using.

Visualizations, particularly interactive ones, will be well-received. That said, do not overuse visualizations. You may be better off with one complicated but well-crafted visualization as opposed

to many quick-and-dirty plots. Any plots should be well-thought out, properly labeled, informative, and visually appealing.

The R code is there to support the technical reader who wishes to dig into your work – not to substitute for written explanation. Do *not* present long unbroken chunks of R code without offering written explanations. Part of the rubric will be code quality: is it straightforward for others to follow.

I will be reproducing your analyses so please be sure that the process is clearly laid out in the GitHub repo README.md file (see repo evaluation below).

1.8 Group Dynamic Report

Ideally, all group members would be equally involved and able and committed to the project. In reality, it doesn't always work that way. I'd like to reward people fairly for their efforts in this group endeavor, because it's inevitable that there will be variation in how high a priority people put on this class and how much effort they put into this project.

To this end, I would like each of you (individually) to describe how well (or how poorly!) your project group worked together and shared the load. This should be submitted via an individual DM to me on Slack by the end of the day on **Tuesday, May 14th**. (If you worked individually, you do not need to submit this report.) Please provide some specific comments describing each member's overall effort. Was there a group member who really put out exceptional effort and deserve special recognition? Conversely, was there a group member who really weren't carrying their own weight? And then, at the end of your assessment, estimate the percentage of the total amount of work/effort done by each of you. (Be sure your percentages sum to 100%!)

For example, suppose you have group members: X and Y. In the event that each member contributed equally, you could assign:

- 50.0% for member X, 50.0% for member Y

Or in case person X did twice as much work as the other member, you could assign:

- 66.7% for member X, 33.3% for member Y,

I will find a fair way to synthesize the (possibly conflicting) assessments within each group and fairly incorporate this assessment of effort and cooperation in each individual's overall grade. Don't feel pressured to share your reports with one another. Just be fair to yourselves and to one another. Let me know if you have any questions or if you run into any problems.

2 Assessment Criteria

Your project will be evaluated based on the following criteria:

- Originality/Interest: Is the topic original, interesting, and substantial – or is it trite, pedantic, and trivial? How much creativity, initiative, and ambition did the group demonstrate? Is the basic question driving the project worth investigating, or is it obviously answerable without a data-based study?

- Degree of Difficulty: How challenging was the project? Were the data particularly large, complex, and/or messy? Did the data come in an obscure format? Was a scraper or API necessary to acquire the data? Was a challenging visualization or applet constructed? Were any elements from outside the coursework necessary to complete the project?
- Design: How well were the graphical elements of the project designed? Were they clunky or elegant? Was a truly original view of the data presented? Were any interactive elements usable?
- Meaning/Analysis: Did we learn anything meaningful from this project? Are the chosen analyses appropriate for the variables/relationships under investigation, and are the assumptions underlying these analyses met? Are the analyses carried out correctly? Did the group make appropriate conclusions from the analyses, and are these conclusions justified?
- Write-Up and Code: How effectively does the write-up communicate the goals, procedures, and results of the study? Are the claims adequately supported? Does the writing style enhance what the group is trying to communicate? How well is it edited? Does the code style and naming reflect best practices? Are text and analyses effectively interwoven? Clear writing, correct spelling, and good grammar are important.
- Oral Presentation: How effectively does the oral presentation communicate the goals, procedures, and results of the study? Do the slides help to illustrate the points being made by the speaker without distracting the audience? Do the presenters seem to be well-rehearsed? Did they properly budget their time? Do they appear to be confident in what they are saying? Are the arguments persuasive?

2.1 Presentation Evaluation

The form below is an example of what I will use to evaluate your oral presentations.

- Volume: Are the speakers loud enough? Can you hear them even in the back of the room?
- Presence: Do the speakers command the room? Do they draw your attention or do they just look like they want this to be over as soon as possible?
- Emphasis: Do the speakers use intonation to signal what words or concepts are most important? Or do they appear to be reciting a speech flatly?
- Eye Contact: Are the speakers engaged with the audience? Do they seem connected or do they seem to be going through the motions?
- Clarity: Did the speakers convey a clear sense of what the project is about? If you had to tell someone what their project was about in three sentences could you do it?
- Mastery: Do the speakers appear to have full command of the material? Were there obvious gaps in their knowledge of the subject matter? Were they able to answer questions coherently?

2.2 Repo Evaluation

Your group project GitHub repo will be evaluated based on the following criteria:

- A README.md detailing all the steps to reproduce your results. I will use the readme to attempt to reproduce your results.
- No references to local file structure. All references should be relative (not absolute) path-names.
- Appropriate naming of files and objects.
- Consistent code style and appropriate formatting.
- Helpful comments in code.
- No scrap code or files left in the repo. No passwords or personal information.

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